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Describing Variations in the Carapace Shape of the Red-Clawed Crab *Perisesarma Bidens*

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ABSTRACT

Carapace shape is considered an important taxonomic character in crustaceans. Dimorphism was reported in many species and whether the variations between sexes are found in crabs, this study was therefore conducted. The variations in the carapace shape between male and female sexes of the red-clawed mangrove crab, Perisesarma bidens was investigated using geometric morphometric techniques involving tools in imaging, geometry, statistics and biology. Thin-plate spline deformation grids showed that the deformations in males is primarily an increase in antero-posterior length and width while females showed changes which vary from narrowing to broadening of the frontal, caudal and carapace width. Relative warp analysis showed insignificant variations in the different regions of the carapace within sexes. Euclidean Distance Matrix Analysis (EDMA) showed increased antero-posterior length of the male and broadening of the posterior in females. The broadened posterior margins of females observed here maybe linked to functional aspects of reproduction while the 50:50 misidentification of the male carapace shape can be due to differences in growth, clutch size, to reproductive behavior and agonistic encounters. Results of Canonical Variate analysis suggested the absence of significant variation in carapace shape between sexes. Further, discriminant analysis showed the absence of sexual dimorphism in the carapace shape of P. bidens due to high misclassification of males unlike the females. The results of the study therefore clearly show the importance of utilizing geometric morphometric tools in quantitatively describing variations in organisms.

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INTRODUCTION

Analysis of forms in living organisms evolved from basic biometric morphology [1], functional morphology [2], structural morphology [3], and theoretical morphology [4]. Many studies on crustaceans' body shapes made use of traditional morphometrics by applying bivariate or multivariate methods to sets of variables such as distances angles, ratios, etc. between points from the animal body. With the advances in imaging, statistics, biology and mathematics however, have resulted in a new field of shape analysis called geometric morphometrics (GM) in which "landmark coordinates" are recorded in several dimensions to directly capture the geometry of the object [5]. GM have emerged as method for comparing organisms' shapes allowing a quantitative study of shape variation and the identification of its causes [6]. The generated set of shape variables in GM can be used to test statistical hypotheses providing a means of visually describing patterns of shape differences in the data [7]. Some of the applications of GM in studying crabs' rigid exoskeleton are for taxonomic identifications, fishery stocks, maturity instars, ontogenetic stages, or sexual dimorphism [8-9] . Sexual shape dimorphism in body shape changes was argued to have important ecological consequences and evolutionary trends [10] thus, was applied in this study on *Perisesarma bidens*, the most common mangrove crab inhabiting intertidal zone to high tide mark, This study will look into describing variations in carapace

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shapes between the male and females sexes which may have important ecological consequences since the species is adapted to the sediment conditions, tidal fluctuations, and varying salinities found in the mangroves [11].



Fig. 1: The mangrove crab, *Perisesarma bidens*

Methodology:
Sampling site, collection and preservation:

The crabs were collected from the mangrove areas of Barangay Dalipuga, Iligan City. It is the first coastal barangay of Iligan City from the province of Misamis Oriental. The mangrove areas are patchily distributed along the coastal areas of the barangay. Its geographic coordinates are 8.308110 North and 124.254756 East (Figure 2). The specimen were collected by opportunistic hand picking for a duration of three months from January to March 2014. The crabs were placed in a bucket with minimal seawater and brought to the laboratory, cleaned and photographed using a digital canon camera mounted on a tripod with a fixed focal distance.



Fig. 2: Location map of Barangay Dalipuga, Iligan City, Philippines

Acquiring Landmark Data by Geometrics Morphometrics:

Geometric morphometric data consist of 2D Cartesian landmark coordinates. Landmarks are points of correspondence on each specimen that match between and within populations or, equivalently, biologically homologous anatomical loci recognizable on all specimens in the study [12].

There are 22 landmark points for the carapace (Fig.3). Landmarks are found in all species in order to capture the general shape of the carapace in all individuals. These landmarks are homologous in all individuals. These landmarks were digitized thrice by the same researcher on different days with TPS Dig2 software, version 2.16 [13]; available in http://life.bio.sunysb.edu/morph). . The software produces corresponding values to x and y coordinates in each designated landmark points. The X and Y values were saved in Microsoft Excel for superimposition procedure.A Generalized Procrustees Analysis (GPA) was performed for each carapace to separate the size and shape components of the landmark configuration. GPA was applied by representing landmark configurations by a centroid (the center of mass of a configuration), by calculating the size of the centroid of each configuration for the value one, and by then rotating the landmark configuration to obtain a least squares fit to the corresponding anatomical landmarks [7].

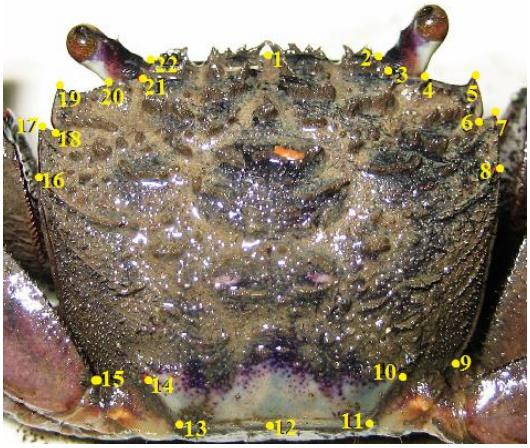


Fig. 3: Assignment of the 22 landmark points on the dorsal surface of the carapace.

Table 1: Landmark points on the dorsal surface of the carapace of *Perisesarma bidens*.

Landmark No.	Description	Landmark No.	Description
1	anterior midgroove	12	dorsal midpoint
2	proximal endpoint of right orbit	13	left dorsal endpoint
3	between 2 and the midpoint of right orbit	14	the notch between the left dorsal endpoint and the left anterolateral border
4	between the midpoint of the right orbit and the exorbital angle	15	left anterolateral border
5	right exorbital angle	16	base of the first left anterolateral tooth
6	base of right exorbital angle	17	tip of the first left anterolateral tooth,
7	tip of the first right anterolateral tooth	18	base of left exorbital angle
8	base of the first right anterolateral tooth	19	left exorbital angle
9	right anterolateral border	20	between the left exorbital angle and the midpoint of the left orbit
10	the notch between the right anterolateral border and the right dorsal endpoint	21	between the midpoint of the left orbit and the proximal endpoint of the left orbit
11	right dorsal endpoint	22	proximal endpoint of the left orbit

Superimposition Procedure:

Partial procrustees superimposition is the most widely used superimposition method and forms the basis for many subsequent operations in geometric morphometrics. Removal of differences in location is achieved by centering configurations and this is done by calculating the centroid of each configuration, and then making the centroid the origin of a new ordinate system. Removal of differences in size between configurations is achieved by rescaling each configuration so that they share a common centroid size. Removal of differences in orientation between two configurations is achieved by rotating one configuration around its centroid until it shows minimal offset in location of its landmarks relative to other configuration. The mismatch in landmark locations between the target and reference configurations is quantified as the squared distance between their corresponding landmarks , summed over all landmarks: the square root of this is the partial procrustees distance between those configurations [14] .

Thin-plate splines:

The thin-plate spline [12] is a mathematical model for producing a smooth grid depicting how one shape (the reference form) can be deformed into another (the target form). Warp scores, convenient variable used in many statistical analyses, can be generated through mathematical decomposition of the spline.

Relative warps:

The relative warps were viewed as the principal components of the set of thin-plate transformations from the mean shape to each of the shapes under study. Relative warps produced the principal warps that describe shape deformations of the reference configuration at different spatial scales. The relative warps were visualized with the thin-plate spline transformation grids. An increase or decrease of the amplitude factor away from zero, the original landmark configuration and grid progressively deformed according to the selected relative warp. Relative warp scores were shown in numerical matrix and the pairs of consecutive relative warps were shown in scatter plots. The relative warps are the principal components of the variation among specimen in the space of the principal warps [12]. Each relative warp explains a percentage of the overall variation in decreasing order, as with principal components [6]. Relative warps was used in this study to significantly ordinate set of shapes.

Euclidean Distance Matrix analysis:

The data analyzed are x and y coordinates of the 22 landmark points of the dorsal carapace of the red-clawed crabs. The raw coordinate data was subjected to Euclidean Distance Matrix Analysis (EDMA) to measure all distances between landmarks. The form matrix does not change and transforms landmark data into a data matrix with a set of distances between all pair of landmarks [15]. EDMA determines the loci and magnitude of sexual dimorphism in *P. bidens*.

Principal component analysis:

To determine the occurrence of sexual dimorphism in the carapace shape, principal component analysis (PCA) was performed on the variance-covariance matrix of the residuals from the GPA. The principal component scores were used as new variables to characterize the shape. This approach allowed the scores to be used as independent variables and served to reduce the dimensionality of the data [16].

Discriminant analysis/Hotelling's test:

Differences in shape between the sexes were likewise tested with a discriminant analysis. This analysis computed the classification percentages and performed a cross-validation between the groups [16]. Discriminant analysis (DA) is a very useful tool for classifying cases into different groups with a better chance of accuracy, and for detecting the variables that will allow the researcher to discriminate between different groups. Discriminant analysis confirms or rejects the hypothesis that the two groups are morphologically distinct.

Canonical variate analysis:

CVA is a method of multivariate analysis in which the variation among groups is expressed relative to the pooled within-group covariance matrix. Canonical variate analysis finds linear transformations of the data which maximize the among-group variation relative to the pooled within-group variation. The canonical variates then may be displayed as an ordination to show the group centroids and scatter within groups. This may be thought of as a "data reduction" method in the sense that one wants to describe among-group differences in few dimensions. The canonical variates are uncorrelated, however the vectors of coefficients are not orthogonal as in Principal Component Analysis. The method is closely related to multivariate analysis of variance (MANOVA), multiple discriminant analysis, and canonical correlation analysis. A critical assumption is that the within-group variance-covariance structure is similar, otherwise the pooling of the data over groups is not very sensible [17].

Principal component analysis, canonical variate analysis and discriminant analysis were done using PAST (Paleontological Statistics package for Windows version 2.0) [15].

RESULTS AND DISCUSSIONS

Visualization of the carapace shape using the Thin-Plate Spline Deformation Grids show *P. bidens* has enlarged or reduced regions relative to the others. Deformation grids show that the deformations of the male carapace is primarily an increase in the antero-posterior (AP) length and width (Fig. 4). It can be seen from the figure that the mean shapes of the male and female crabs show some degree of variations in the left and right sides including the posterior region of the carapace. Females show changes which vary from narrowing to broadening of the frontal , caudal and carapace width.

Within sexes, the variations are described graphically in Figure 5 and qualitatively in Table 2. Relative warp analysis show that the percentage variations in shape ranges from 8.38-30.08% for female and 5.2-42.31%

for male suggesting that the variations in the different regions of the carapace within the male and female populations is insignifiant. Likewise, relative warp analysis show that the deformations in the male carapace primarily involves increase in antero-posterior (A-P) length as well as broadening of the anterior while the dominant deformation in female red-clawed crabs is broadening of the posterior region of the carapace.

Quantitative analysis using Euclidean Distance Matrix Analysis (EDMA) show 13 interlandmarks describing the variations in the carapace (Figure 7, Table 3). EDMA revealed increased antero-posterior (A-P) length of the male carapace, and less-pronounced increase in width and changes in the frontal region. These are due to the following deformations: (1) bilateral expansion of the midlateral margins of the carapace; (2) expansion in the frontal region; (3) compression in the anterolateral margins; and (4) expansion of the posterior half directed posteriorly. For the female, there was broadening of the caudal region of the female carapace due to increased interlandmark distances between the anterolateral and posterolateral margins secondary to the following deformations: (1) expansion of the caudal landmarks ; (2) bilateral expansion of the middle to posterior regions; (3) lateral expansion of the right middle to posterolateral region ; (4) expansion of the right anterolateral margin directed anteriorly/forward; and (5) simultaneous lateral expansion of the caudal landmarks and of the right anterolateral margin (Table 3). Likewise, LM 9 and 10, 14 and 15 (Figure 12B) which correspond to the posterolateral regions account for the highest variation in the carapace shape of female crabs. Primarily, deformations due to expansions in these regions bring about broadening of the caudal region of the carapace.

To determine if the observed variations differ between sexes, Canonical Variate Analysis (CVA) was conducted (Tables 4 and 5). CVA was used to test for significant differences in carapace shape between the two sexes and to determine the confidence with which *a priori* specimen classification is supported by shape data. The resulting "multivariate F" measures the heterogeneity among a set of multivariate centroids with respect to the pooled within-group covariance matrix for the observed correlations among variables show the absence of sexual dimorphism in *P. bidens* (Table 3). Statistical results of this present study show Wilk's lambda = 0.7996; Pillai Trace = 0.2004; and P(same) = 0.7294 suggest that there is no significant variation in the carapace shape between sexes of *Perisesarma bidens.* While females were 87% correctly classified, the males show an almost 50:50 reclassification thus correct classification of males is difficult (Table 4).

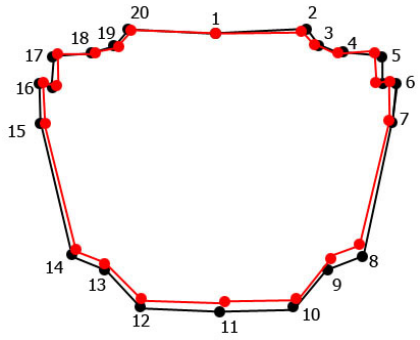


Fig. 4: Mean shapes of *Perisesarma bidens* (red: male; black: female).

Table 2: Qualitative descriptions of shape variations of the male and female carapace based on relative warp scores.

Relative Warp	MALE	FEMALE
1	Expansion at LM 15-22 causing dorsal bending of the right anterolateral region; compression at LM points 5-10 causing diagonal bending across LM points 3 and 13 ; the compression results in ventral bending of the left and dorsal bending of the right lateral regions of the carapace; narrowing of the left lateral border.	Expansion at LM points 10-12 causing dorsal bending of the posterior region and subsequent ventral bending at the upper left diagonal half due to compression of LM points 21, 22 , 1,2, 3 and 4 directed towards the midline. Overall result is decrease in width of the anterior and decrease in width of the posterior.
2	Expansion at the anterior LM points causing stretching of the carapace through the midline and subsequent ventral bending of the anterior part resulting in an increased anteroposterior length.	Dorsal expansion at the right lateral half at LM points 13-16 postero-medially and compression of the left lateral half most pronounced in LM points 2-7 of the anterolateral spines resulting in dorsal bending of the anterior and right lateral half and ventral bending of the anterolateral part.
3	Expansion at the midanterior points 22, 1 and 2 and midposterior LM points 11, 12 and 13 minimally stretching the middorsal carapace; compression at the right and left anterolateral and midlateral regions corresponding to LM points 4-10 and its opposite points.	Expansion at LM points 4-7 of the anterolateral spines; Compression more pronounced at LM points 22, 1 and 2 of the anterio-median part , less pronounced at LM points 11, 12 and 13 of the posterior region resulting in shortening of the carapace width. Overall result is decrease in width most pronounced in the anteriorpart of the carapace.
	Expansion of both right and left midlateral regions at LM 5-9 and opposite landmarks directed posteriorly resulting	Expansion at LM points 6-9, 15-1directed left posterolaterally, less pronounced at LM point 11;

4	to (1) increased anteroposterior length of the lateral parts only , and (2) compression at LM 12-14 resulting to dorsal bending of the right posterior half of the dorsal carapace.	compression at LM points 12-14. The overall result is increase in width of the carapace due to expansion in 2 regions.
5	Expansion of the posterior half, most pronounced at both midlateral parts, as well as expansion of the the midanterior regions towards the anterior , stretching the carapace anterior; compression in both anterolateral regions at LM points 3-8 and opposite points bending ventrally the anterolateral spines ventrally, the midanterior dorsally. The result is increased anteroposterior length.	Expansion at LM points 8-16 corresponding to the posterior half of the dorsal carapace and the anterior median dorsal carapace directed anteriorly. The expansion produces compression and ventral bending of the anterior half of the carapace. Overall result is increase in antero-posterior length of the carapace.
6	Expansion at right anterolateral spines, 16-2 and 11-13; compression at the anterior half, LM 5-8; increase width.	Expansion at LM 11-13 of posterior region, 3-5 and 18-21 of the right and left anterolateral spines; compression most pronounced at LM 13-15 causing ventral bending of right posterolateral part; decrease both A-P and width.

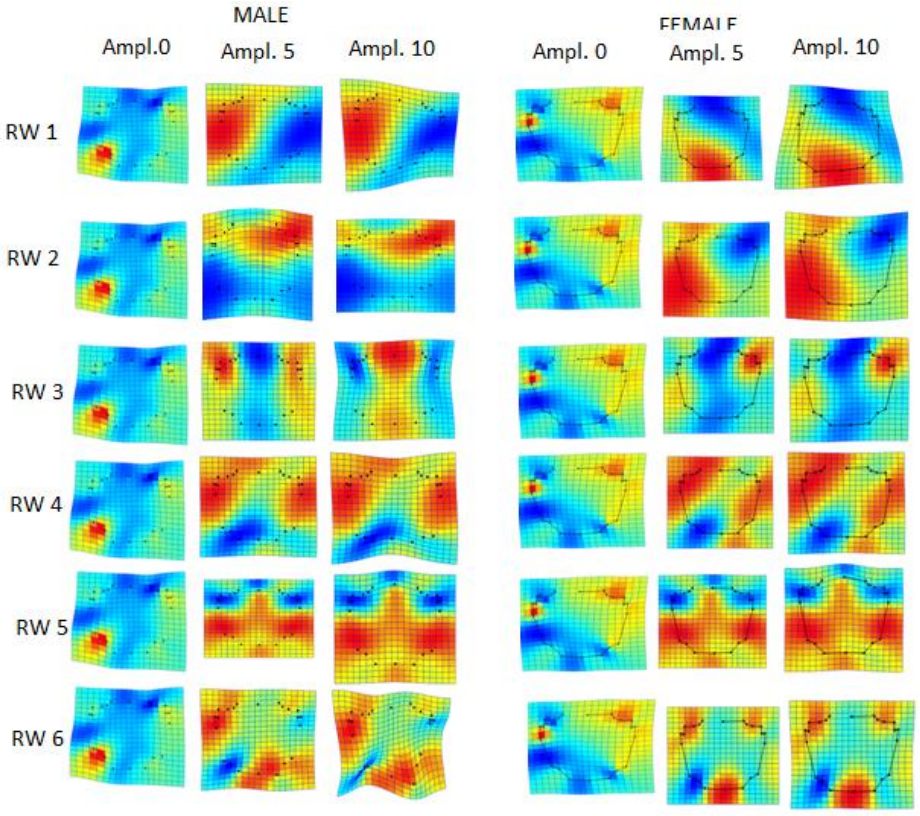


Fig. 5: Thin-plate spline visualization showing carapace shape deformations of the male and female *Perisesarma bidens* at amplitudes 0, 5, and 10 based on relative warp scores.

Table 3: Interlandmark distances with the highest PCA loading values

FEMALE		MALE	
Interlandmark Distances	Values	Interlandmark Distances	Values
9-19	0.467995671	5-15	0.467656118
9-17	0.464656038	9-19	0.472865700
5-15	0.459180650	9-17	0.467191420
7-17	0.454574732	7-15	0.466876458
7-15	0.454574732	9-18	0.452874771
9-16	0.454539315	7-16	0.451470110
8-17	0.453854942	5-13	0.451087999
9-18	0.451114477	8-17	0.449943458
5-16	0.446761918	8-16	0.448657419
8-16	0.446739520	8-15	0.446786093
11-19	0.446026845	7-17	0.446475302
8-19	0.440943604	5-14	0.440513243
8-18	0.438755866	5-16	0.438528035

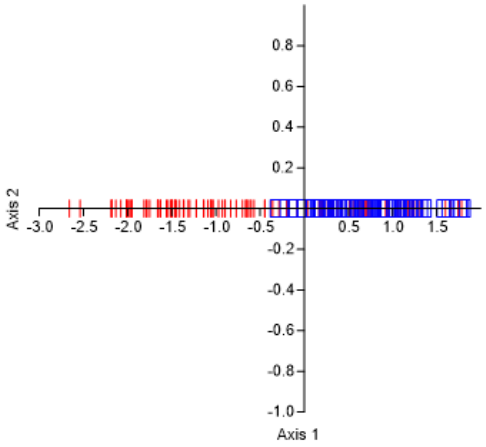


Fig. 6: CVA scatter plot of the carapace landmarks showing the first two significant components with the highest percentage of variance.

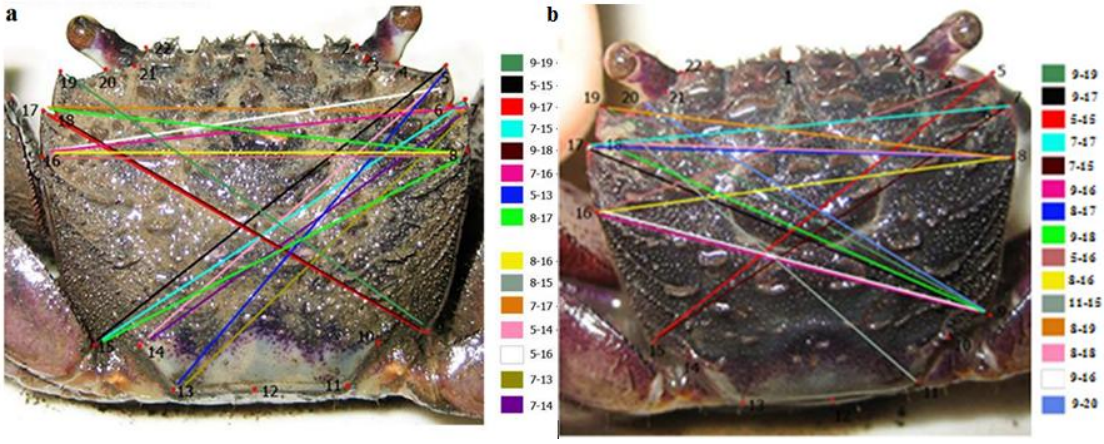


Fig. 7: Interlandmark distances with the highest PCA loading values for the significant components of the landmark coordinates of the (a) male and (b) female carapace.

Table 4: Canonical variate analysis of the Relative warp scores between sexes of *P. bidens.*

Wilk's Lambda = 0.7996	Pillai Trace = 0.2004
df1 = 44	df1 = 44
df2 = 147	df2 = 147
F = 0.8374	F = 0.8375
P(same) = 0.7493	P(same) = 0.7492

Table 5: Confusion matrix between male and female crabs

	Male	Female	Total
Male	76	54	130
Female	8	54	62

The results of this study have clearly shown the distinctness of the female carapace of *P. bidens* but not the males (Table 5). The results is in contrary to those obtained in studies of other crab species such as those of *Perisesarma sp*., *Aegla neuquensis* and *Aegla riolimayana* [18]; in Aeglidae [19] ; in *Liocarcinus depurator* [20]; and in *Portunus pelagicus* [21]; *Necora puber* (L.) and *Bathynectes maravigna* (Costa) [22] and in many species of Decapoda [23] ; [16,18, 24, 25 ,]. In these studies female individuals have broadened posterior margin of the carapace related to abdomen base while the male have broadened posterolateral region [26]. However, the observed broadened posterior margin of the carapace of the females in *P. bidens* can be related to the abdomen base for greater relative volume for gonad development and external surface for carrying eggs on the abdominal pleopods [20, 27]. The almost 50:50 misidentification of the male carapace shape in *P. bidens* can be differences in growth [18, 20, 28] and maybe clutch size [28] or to reproductive behaviour and agonistics encounters in males [20].

Conclusion:

This study generally show the absence of sexual dimorphism in carapace shape in *P. bidens* due to high misclassification of males unlike the females. Nevertheless, the variations within the sexes can be quantitatively described using the tools used in geometric morphometric analysis. The results can be more biologically meaningful if the study will be coupled with functional hypotheses regarding the adaptive significance of differences in body shape such as variations in relation to feeding behavior, sexual selection and reproductive modes could contribute to a better understanding of adaptive functions related with the body structures of this species.

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